CSIE 2344, Spring 2026 — Laboratory 2 Report

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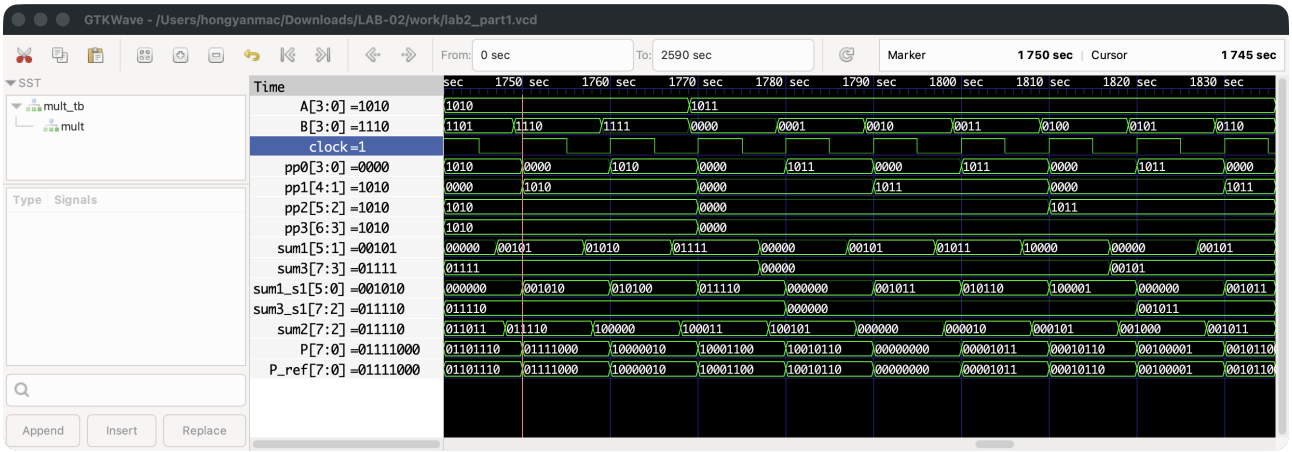
Part 1 (32 pts)

Q1. (12 pts) Module mult_fast

The completed Verilog module mult_fast with proper indexes filled in:

```
// pipelined fast multiplier
module mult_fast(
    output reg[7:0] P,  // product
    input[3:0] A, B,    // multiplicand and multiplier
    input clk           // clock (posedge)
    );
    // stage 0 (input)
    reg[3:0] a_s0, b_s0;
    always @(posedge clk) begin
        a_s0 <= A;
        b_s0 <= B;
    end
    // stage 1
    wire[3:0] pp0 = a_s0 & {4{b_s0[0]}};
    wire[4:1] pp1 = a_s0 & {4{b_s0[1]}};
    wire[5:2] pp2 = a_s0 & {4{b_s0[2]}};
    wire[6:3] pp3 = a_s0 & {4{b_s0[3]}};
    reg[5:1] sum1;
    always @(pp0, pp1)
        sum1[5:1] <= #7 pp0[3:1] + pp1[4:1];   // 4-bit adder
    reg[7:3] sum3;
    always @(pp2, pp3)
        sum3[7:3] <= #7 pp2[5:3] + pp3[6:3];   // 4-bit adder
    reg[5:0] sum1_s1;
    reg[7:2] sum3_s1;
    always @(posedge clk) begin
        sum1_s1 <= {sum1, pp0[0]};
        sum3_s1 <= {sum3, pp2[2]};
    end
    // stage 2 (output)
    reg[7:2] sum2;
    always @(sum1_s1, sum3_s1)
        sum2[7:2] <= #8 sum1_s1[5:2] + sum3_s1[7:2];   // 6-bit adder
    always @(posedge clk) begin
        P <= {sum2, sum1_s1[1:0]};
    end
endmodule
```

Q2. (12 pts) Waveform with lab2.sav (1750 ~ 1800 sec)



The waveform shows the pipelined multiplier operating with a 10-tick clock cycle. Each stage's signals (a_s0/b_s0, pp[0..3], sum1/sum3, sum1_s1/sum3_s1, sum2, P) progress correctly through the three pipeline stages, and the output P matches the reference P_ref, confirming functional correctness.

Q3. (8 pts) Latency

Latency is the worst-case time from a steady input until the final-stage register captures the corresponding product. Timing breakdown:

```
t = 0    Input A, B becomes steady (just after the previous posedge clk).
t = 10   Stage-0 register latches A, B  → a_s0, b_s0.
t = 17   sum1, sum3 settle after the 4-bit adder delay (#7).
t = 20   Stage-1 register latches sum1_s1, sum3_s1.
t = 28   sum2 settles after the 6-bit adder delay (#8).
t = 30   Stage-2 register latches the final product P.
```

```
Total latency = 30 ticks (= 3 clock cycles).
```

Answer: 30 ticks.

Part 2 (16 pts)

Q1. (8 pts) Minimum clock cycle

The minimum clock period is bounded by the slowest combinational path between consecutive pipeline registers:

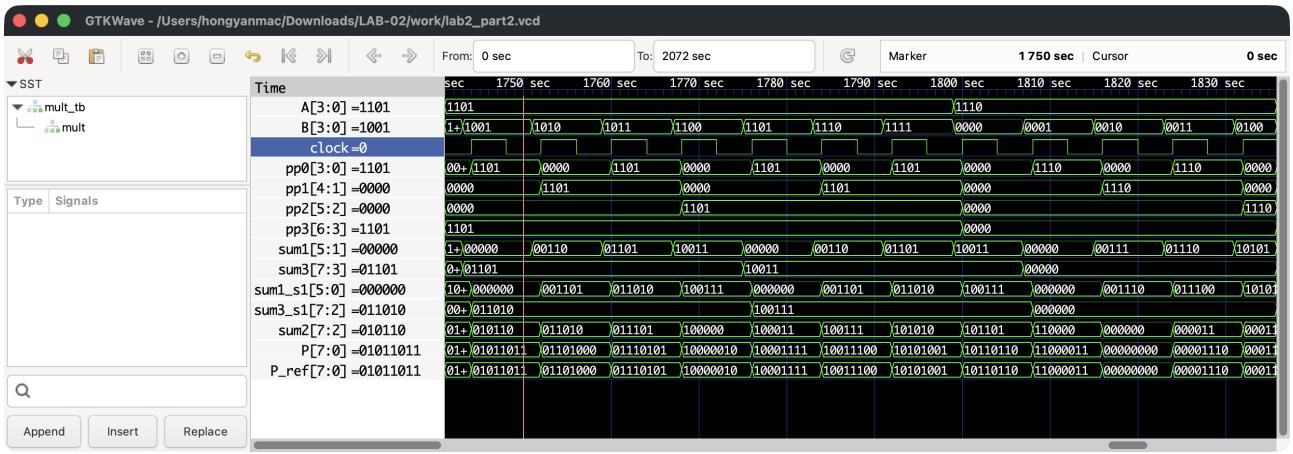
```
Stage 0 → Stage 1 : 4-bit adder delay  = 7 ticks
Stage 1 → Stage 2 : 6-bit adder delay  = 8 ticks   ← bottleneck

T_min = max(7, 8) = 8 ticks.
```

To realise this, mult_tb is modified so the clock half-period becomes 4 (#4 clock <= ~clock;), the input-update interval becomes 8 (#8), and P_ref is delayed by two clock cycles (#16 A*B). All 256 input combinations are exercised and P matches P_ref throughout the simulation — no "Product is wrong" messages.

Answer: 8 ticks.

Q2. (8 pts) Waveform with lab2.sav (1750 ~ 1800 sec)



With the clock period reduced to 8 ticks, the waveform is denser: each 8-tick period contains one full posedge/negedge cycle. Pipeline timing remains valid because each stage's combinational delay (7 ticks for 4-bit adders, 8 ticks for the 6-bit adder) fits inside the new clock period, and P continues to track P_ref correctly.